EXERGYNET
Persistent Machine Infrastructure

Memory Grows. Context Does Not Have To.

ExergyNet: Persistent Machine Infrastructure
for the 100M-Token Operating Horizon

Seven Ezumba
Chief Architect, ExergyNet

Veena Bontu
Co-author

October 2026

EXTERNAL PUBLIC-SAFE · PROPRIETARY IMPLEMENTATION WITHHELD

Validated outcomes are reported within their tested operating envelopes.
Proprietary implementation details are intentionally excluded.

exergynet.org

1. Executive Summary

Persistent machine intelligence needs memory that keeps growing without forcing the model to carry it. ExergyNet is independent infrastructure that holds persistent state, evidence, identity and authority outside the model and outside the accelerator, so a bounded amount of context reaches the model however much state accumulates.

At a validated 100M-token operating point, the model received a median of 800 tokens of active context. Retrieval at that scale ran a median 220.5× faster per query, with no right-to-wrong answer changes across 190 matched queries.

≈101M	800	220.5×	415.9×
tokens of persistent state (validated development envelope)	median active context tokens at 100M	median paired retrieval speedup	P95 paired retrieval speedup

Result	Headline	Tested envelope
100M operating point	≈101M est. tokens; median active context 800; 190/190 executions completed	Development cohort, single run
Retrieval at 100M	220.5× median / 415.9× P95 paired speedup; 0 right-to-wrong; 0-token median context change	190 matched queries
Scaling	Active context stayed bounded from 32K to ≈101M tokens; no statistically significant positive trend	12-point development campaign
Physical state economics	Whole-query GPU energy rose ≈81× from 1M to 100M; model-window GPU energy did not scale with it	Matched 50-query cohort, four scales
Accelerator throughput	≈11.3× correct-task throughput vs full-context replay	Tested H200 workload
Cross-accelerator	Advantage measured on both NVIDIA GPU and Google TPU	Respective tested workloads
Authority	Deterministic layer rejected an action the model had accepted	Live validation, reproduced

2. The Persistent-State Problem

The first generation of model deployment was stateless: a prompt arrived, an answer left, nothing remained. Persistent agents change that. They accumulate decisions, evidence, sessions and grants of authority, and that history has to be available the next time they act.

The common response is to put the history back into the prompt. Then accumulated state and active context become the same resource: the more an agent remembers, the more every inference costs, until the model's context limit is reached and the history no longer fits.

ExergyNet separates the two. **N** is the persistent state an agent has accumulated. **K** is the context actually delivered to the model for one query. The question this paper measures is whether K must

grow as N grows. Within the tested envelope, it did not.

3. What ExergyNet Is

ExergyNet is infrastructure beneath persistent machine intelligence and above heterogeneous compute.

> **Models and accelerators may change. Persistent state and authority infrastructure should not have to.**

ExergyNet provides	ExergyNet is not
Persistent state and evidence that outlive any single model or session	A model
Machine identity and deterministic execution authority	A feature of any one accelerator or vendor stack
Resource accounting for what intelligence consumes	A retrieval add-on or a RAG implementation
Model-agnostic operation across GPU and TPU environments	Tied to one accelerator generation

4. 100M Validation

What we measured. Active model-facing context (K) per query as the persistent corpus grew from 32,000 to about 101 million estimated tokens, on an adversarial synthetic benchmark, with an NVIDIA Nemotron-class reasoning model on NVIDIA A100 GPUs.

Metric at the 100M operating point	Value
Persistent state	101,152,694 estimated tokens (3,161× the 32K baseline)
Executions completed	190 / 190, zero errors
Median active context (K)	800 tokens
Mean / P95 active context	806 / 891 tokens
Accuracy on the development cohort	45.3%
State-to-active-context ratio (derived)	≈125,000 : 1

The ratio divides corpus size by mean active context. It is a derived figure, not a compression ratio, and does not mean every query touched every token.

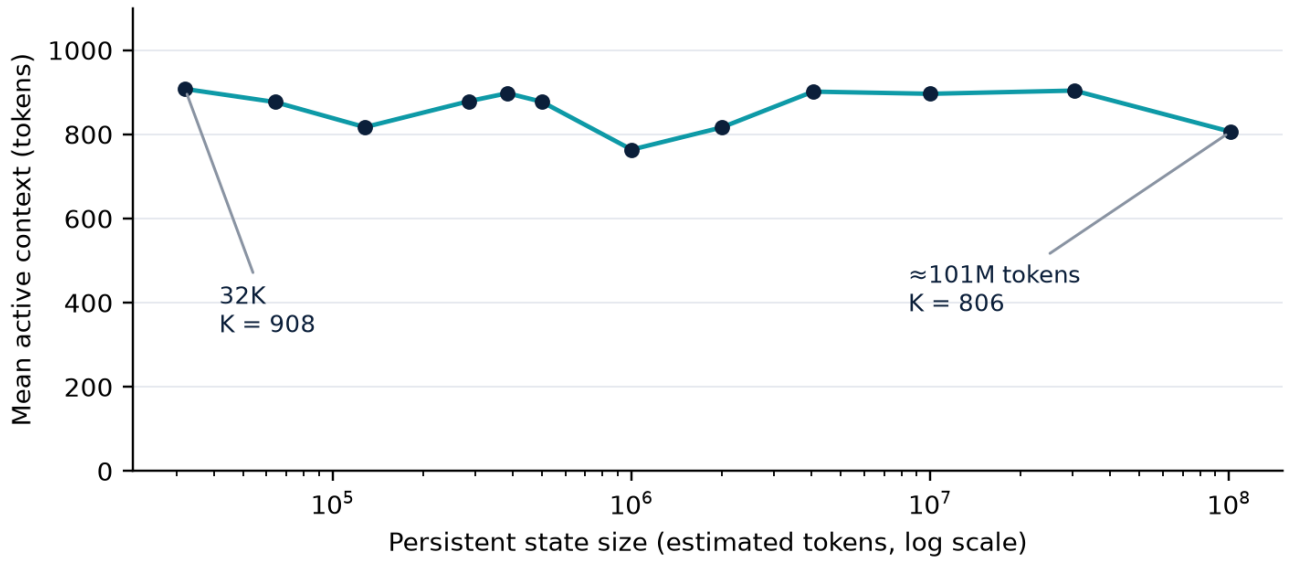


Figure 1. Mean active context at twelve measured scale points, 32K to ≈101M tokens.

Result. Across more than three orders of magnitude of persistent-state growth, active model-facing context remained bounded within the tested campaign. A twelve-point regression of K against corpus size did not detect a statistically significant positive trend (slope -4.3×10^{-7} tokens per corpus token; 95% interval -1.5×10^{-6} to $+6.9\times10^{-7}$, which includes zero). A separate sealed genuine holdout at 10.1 million tokens (950 executions, zero errors) returned a mean active context of 896 tokens.

Why it matters. The model's working set stayed the same size while the state behind it grew by about 3,000×. Accelerator cost per query is set by what the model reads, not by what the system remembers.

5. Retrieval Performance at 100M

What we measured. Per-query retrieval time at 100M tokens for the same 190 queries, before and after an enhanced retrieval configuration, with the model, prompts and scoring held identical.

220.5×	**415.9×**	**0**	**0 tokens**
median paired speedup	P95 paired speedup	right-to-wrong changes in 190	median active-context change

Metric (190 matched queries)	Result
Paired per-query retrieval speedup, P25 / P50 / P75 / P95	215.6× / 220.5× / 237.8× / 415.9×
Median active-context change	0 tokens (mean −0.5)
Answers correct → incorrect	0
Answers incorrect → correct	5 (accuracy 45.3% → 47.9%; not statistically significant)
Errors	0 of 190

Result. Validated within the tested 100M envelope. Speedups are per-query paired ratios on a development cohort in a single run. This is a measured result at one scale, not a claim of constant-time retrieval.
Why it matters. At 100M tokens the baseline retrieval time was long enough to dominate each query. A speedup of this size at unchanged answer quality moves persistent state from a batch-scale operation toward an interactive one.

6. Physical State Economics

What we measured. GPU energy per query at four corpus scales (1M, 10M, 30M and 100M tokens) on the same matched 50-query cohort, using GPU energy counters on four NVIDIA A100 GPUs, split into the model window and the whole query.

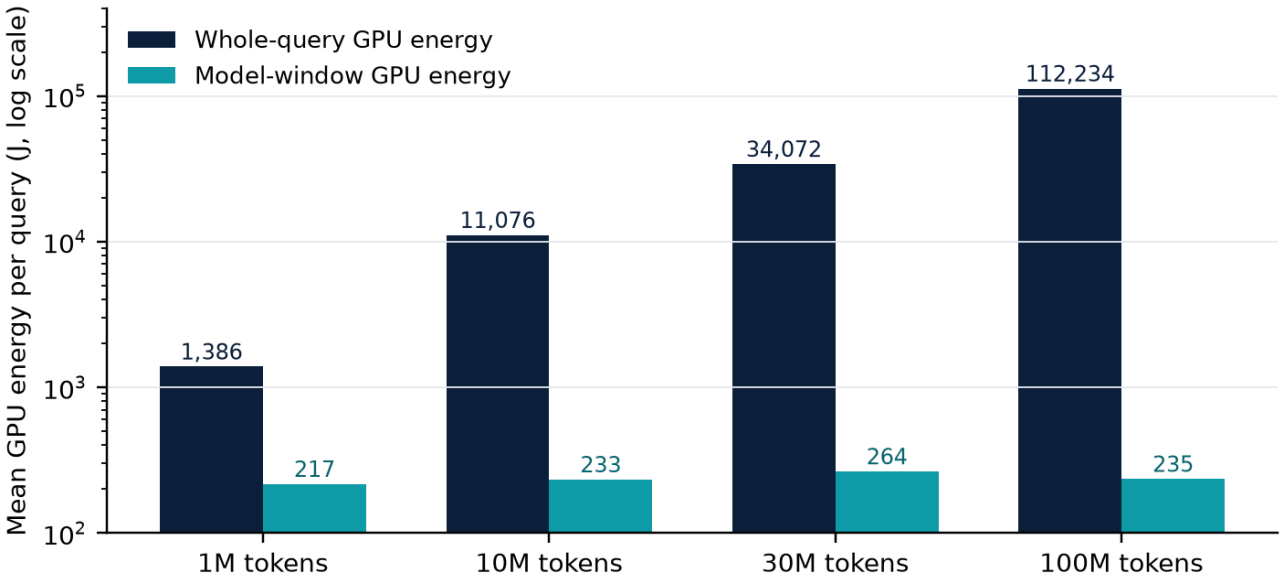


Figure 2. Mean GPU energy per query, baseline retrieval configuration. GPU energy only; host CPU energy is not included.

Corpus	Mean retrieval time	Model-window GPU energy	Whole-query GPU energy
1M tokens	4.8 s	217 J	1,386 J
10M tokens	47.0 s	233 J	11,076 J
30M tokens	140.4 s	264 J	34,072 J
100M tokens	466.4 s	235 J	112,234 J

Result. Across matched 1M, 10M, 30M and 100M workloads, whole-query GPU energy increased materially as baseline retrieval latency scaled, while model-window GPU energy did not grow with it: it stayed between 217 J and 264 J at every scale. Baseline retrieval time was consistent with near-linear scaling across the tested 1M to 100M envelope (fitted exponent 0.99). A small model-window increase at the 30M point was not sustained at 100M.
Why it matters. As persistent state grows, the cost is not more model work. It is accelerator time spent waiting on state. Energy figures in this section are for the baseline retrieval configuration; they are not a measurement of the enhanced configuration in Section 5.

7. Accelerator Independence

H200 throughput. On a single NVIDIA H200 with an NVIDIA Nemotron-class reasoning model, same corpus, same questions and same answer checker, only the way memory is delivered changed. At each method's highest tested operating point that met a 15-second p99 latency target, ExergyNet delivered approximately **11.3× the correct-task throughput of full-context replay**. Active prompt size stayed at roughly 660 to 820 tokens as the corpus grew from 8K to 285K tokens, while full-context prompts grew to 67K tokens and were rejected beyond the model's context limit.

GPU and TPU. In a separate controlled test, the same ExergyNet configuration was run on an NVIDIA A100 and a Google TPU v6e with a 7B-class open model and a 20K-token workload, at three concurrency levels. ExergyNet delivered 1.6× to 2.3× the correct-task throughput of full-context replay on every platform and concurrency level tested, with zero errors across all twelve test cells.

Test	Platform	Measured advantage vs full-context replay	Tested workload
H200 throughput	NVIDIA H200	≈11.3× correct tasks per minute	Nemotron-class model; 8K–285K-token corpus; 15 s p99 target
GPU and TPU	NVIDIA A100	1.69× / 1.88× / 2.26× at concurrency 1 / 2 / 4	7B-class model; 20K-token corpus
GPU and TPU	Google TPU v6e	1.60× / 1.69× / 1.93× at concurrency 1 / 2 / 4	7B-class model; 20K-token corpus

The H200 and cross-accelerator results come from different models and workloads and are not combined or compared with each other.

Why it matters. The measured advantage was demonstrated across both NVIDIA GPU and Google TPU environments within their respective tested workloads. The architecture is not tied to one accelerator family.

8. Authority Is Not Capability

ExergyNet separates model capability from deterministic execution authority. A model can be persuaded. An authority layer that does not depend on the model is not exposed to the same persuasion.

What we measured. A live adversarial instruction was sent to a running authority-review environment. **Result.** A deterministic authority layer rejected the adversarial action after the reasoning model had accepted the instruction, and the result was reproduced. **Why it matters.** A reasoning model can fail adversarially without the authority system failing with it. This is a validated outcome in a tested environment, not a claim that all attacks are blocked.

Affiliation. The author is also CEO of Kunfirm Innovative Services LLC (KTX), holder of FAA Exemption No. 26214 for highly automated UAS operations up to the exemption's approved operating limit. KTX and ExergyNet are separate entities. The exemption is a regulatory record for KTX and is not an FAA evaluation of ExergyNet.

9. Commercial Infrastructure Implication

Accelerators are most valuable when they do intelligence computation. When persistent state lives inside the inference path, accelerator time goes to waiting on state and replaying history instead.

Moving persistent state, evidence and authority into infrastructure of their own means:

- **Accelerator time goes to model work.** The model reads a bounded window however large the state behind it.
- **State survives hardware and model generations.** Replacing the model or the accelerator does not require rebuilding what an agent knows or what it is allowed to do.
- **Authority stays deterministic.** Consequential action is gated independently of whether the reasoning model can be trusted.
- **One infrastructure layer across GPU and TPU environments.** Results are measured on NVIDIA A100 and H200 and on Google TPU v6e.

ExergyNet runs today on NVIDIA GPUs with NVIDIA Nemotron-class models. Additional technical detail is available under an appropriate evaluation framework.

10. Tested Boundaries

Every result in this paper is reported within the envelope in which it was measured.

- The 100M results are a validated development operating point on a development query cohort, in a single run. They are not a genuine holdout. The sealed genuine holdout reaches 10.1M tokens.
- The scaling campaign used an adversarial synthetic benchmark and one model family. Results outside the tested envelope are not claimed.
- Retrieval speedups are paired per-query results for the tested implementation and corpus. They are not asymptotic or complexity claims.
- Energy results are GPU energy on four NVIDIA A100 GPUs for the baseline retrieval configuration. Host CPU energy is not included.
- The H200 result applies to the tested workload and latency target. The GPU and TPU results apply to their own tested workload.

11. Validation Summary

Claim	Result	Tested envelope
Persistent-state scale	≈101M est. tokens	100M development operating point
Active context at 100M	Median 800 / mean 806 tokens	190/190 completed; development cohort
N→K scaling	No significant positive trend, 32K→≈101M	12-point development campaign
Retrieval speedup, median	220.5× (paired)	190 matched queries, 100M
Retrieval speedup, P95	415.9× (paired)	190 matched queries, 100M
Answer changes	0 right-to-wrong; median context change 0	190 matched queries, 100M
Whole-query GPU energy, 1M→100M	≈81× increase (1,386 J → 112,234 J)	Matched 50-query cohort, baseline retrieval

Claim	Result	Tested envelope
Model-window GPU energy	217–264 J at all four scales	Matched 50-query cohort, baseline retrieval
H200 correct-task throughput	≈11.3× vs full-context replay	Tested H200 workload; 15 s p99 target
Cross-accelerator	1.6×–2.3× on A100 and TPU v6e	Each platform's tested workload
Authority	Adversarial action rejected after model accepted it	Live validation; reproduced

12. Conclusion

The measured result is simple. Memory grew by about 3,000×. The context the model read did not.

At the 100M operating point, ExergyNet held the model's working set to a median of 800 tokens, accelerated retrieval a median 220.5× with no answer regressions, and showed that the cost of unmanaged state is accelerator time rather than model work. The measured advantage held across NVIDIA GPU and Google TPU environments within their tested workloads.

> **ExergyNet is infrastructure beneath persistent machine intelligence and above heterogeneous compute. Models and accelerators may change. Persistent state and authority infrastructure should not have to.**

Validated outcomes are reported within their tested operating envelopes. Proprietary implementation details are intentionally excluded.